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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP(Honours)/MGU-BCA(HONOURS)
REGULAR EXAMINATION October 2026
Fifth SEMESTER
Discipline Specific Elective (DSE)-MG5DSECSC302-ARTIFICIAL INTELLIGENCE
(2024 ADMISSION ONWARDS)

Duration :2 hours **Maximum Marks:70**

Students should attend at least one question from each course outcome to enhance their overall outcome attainability

Part A
Very Short Answer Questions
Answer **all** questions
Each question carries **2** marks

1. State-space search is a problem-solving technique that explores possible states and actions to find a path from the initial state to the goal state.
2. Knowledge Representation is the process of representing facts, concepts, relationships, and rules in a form that an AI system can use for reasoning and problem-solving.
3. Conjunction (AND).
4. Mutually exclusive events cannot occur simultaneously. Therefore, $P(A \cap B) = 0$, and $P(A \cup B) = P(A) + P(B)$.
5. Union and Intersection.

Part B
Short Answer Questions
Answer any **6** questions
Each question carries **5** marks

6. The development of Artificial Intelligence began with foundations in mathematics, logic, and computing.

- 1943: Warren McCulloch and Walter Pitts proposed a mathematical model of artificial neurons.
- 1950: Alan Turing introduced the Turing Test to evaluate machine intelligence.
- 1956: The Dartmouth Conference, organized by John McCarthy and other researchers, established AI as a research field.
- 1950s–1960s: Early AI programs were developed for theorem proving, problem-solving, and playing games.
- 1970s–1980s: Research progressed in knowledge representation and expert systems, which used specialized knowledge to solve domain-specific problems.

These milestones established the foundations for modern AI, including machine learning, intelligent agents, and automated reasoning.

7. .AI improves agricultural productivity by analyzing data collected from sensors, drones, satellites, and cameras.

- Crop monitoring: AI analyzes images and sensor data to monitor crop growth, soil conditions, and plant health.
- Disease detection: Machine-learning algorithms and image processing identify symptoms of plant diseases and pest attacks at an early stage.
- Precision irrigation: AI determines the water requirements of crops using soil moisture, weather, and crop data.
- Resource management: AI helps optimize the use of fertilizers, pesticides, water, and agricultural machinery.
- Yield prediction: AI analyzes weather conditions, soil characteristics, and historical crop data to estimate crop yields.

Thus, AI supports timely decisions, reduces resource wastage, lowers production costs, and improves agricultural productivity.

8. .Expert knowledge is the specialized knowledge acquired from human experts through experience, training, and practice. In an expert system, this knowledge is represented as facts, rules, and heuristics in a knowledge base.
The inference engine applies logical rules to the available facts to derive conclusions and recommend solutions. It may use forward chaining, which begins with known facts, or backward chaining, which works backward from a goal.

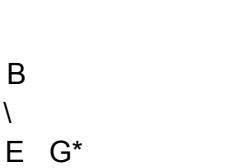
For example, a medical expert system may use symptoms and diagnostic rules to suggest possible diseases. The accuracy of its recommendations depends on the quality, completeness, and reliability of the knowledge base. Therefore, expert knowledge enables an expert system to solve domain-specific problems and support decision-making.

9. Expert systems support environmental management by applying stored environmental knowledge and rules to analyze data and recommend appropriate actions

- Pollution control: Identify possible sources of air, water, and soil pollution.
- Environmental risk assessment: Evaluate potential environmental hazards and their consequences.
- Waste management: Recommend appropriate methods for waste treatment, disposal, and recycling.
- Resource conservation: Suggest measures for efficient use of water, energy, and other natural resources.
- Regulatory compliance: Help assess whether environmental practices conform to applicable standards and regulations.

- For example, an expert system can analyze water-quality measurements and expert-defined rules to identify possible contamination and recommend further investigation or treatment. However, its effectiveness depends on the reliability of the data and the knowledge base.

10. Generate-and-Test: This method generates possible solutions and tests each candidate against the required conditions. It is suitable when candidate solutions can be generated and evaluated easily. For example, it can be used to test possible arrangements in a puzzle.
- Breadth-First Search (BFS): BFS explores nodes level by level, visiting all nodes at one depth before proceeding to the next. It is suitable for finding a shortest path in an unweighted graph. However, it may require considerable memory when the branching factor is large.
- Depth-First Search (DFS): DFS explores one branch as deeply as possible before backtracking to other branches. It is suitable for problems with limited memory requirements and where solutions may occur deep in the search tree. However, it may become trapped in an infinite path unless depth limits or cycle detection are used.
- Thus, Generate-and-Test is appropriate for straightforward candidate checking, BFS for shallowest-solution search, and DFS for memory-efficient deep exploration.

11. .A local heuristic evaluates the immediate state or neighbouring nodes to select the next move. It focuses on the local situation and may not consider the overall cost of reaching the goal.
    A global heuristic considers the broader search problem and estimates how promising a state is in relation to the overall goal or remaining path cost.
- Consider the following search tree:
- 
- Suppose the search begins at S and the goal is G.
- A local heuristic may select A because it appears promising based on its immediate characteristics, even though it does not lead directly to the goal. A global, goal-directed heuristic may favour B because its descendants include G, the goal state.
- Therefore, local heuristics can guide immediate decisions, whereas global or goal-directed heuristic evaluation can help guide search toward the overall solution. The effectiveness of either approach depends on how accurately the heuristic estimates the problem's structure.

12. Inheritance is a knowledge-representation mechanism in which a subclass or individual obtains properties from a more general category in a hierarchy.
    For example:
    - Animal is a general category.
    - Bird is a subclass of Animal.
    - Sparrow is a subclass of Bird.
    - Birds generally have wings and feathers.

If a sparrow belongs to the category Bird, it inherits the properties associated with birds. Since Bird is a subclass of Animal, the sparrow also inherits applicable properties of animals.
Inheritance reduces the need to store the same properties repeatedly and supports efficient knowledge inference. An AI system can derive new facts from existing category relationships. In systems that support default reasoning, exceptions can also be represented, such as the fact that not every bird can fly.

13. Given:$P(D) = 0.01$
P(+| D)=0.98
False positive rate: P(+| D^c)=0.05
$P(D^c) = 1 - 0.01 = 0.99$
Using Substituting the values in Bayes' theorem,
P(D| +)=0.98*0.01 /(0.98*0.01)+(0.05*0.99)]

$$= \frac{0.0098}{0.0098+0.0495}$$

$$= \frac{0.0098}{0.0593} = 0.1653$$

Therefore,
Hence, the probability that the patient actually has Disease D, given a positive test result, is approximately 16.53%

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

14. Expected answer and mark allocation

Answer component	Expected content	Marks
1. Introduction to CSP	Define a CSP using variables X, domains D, and constraints C. Explain that a solution assigns values to variables while satisfying all constraints.	2
2. Role of inference	Explain how inference derives consequences from constraints and reduces the search space before or during backtracking search.	2
3. Constraint propagation	Explain how restricting one variable's domain can lead to the removal of inconsistent values from related variables.	3
4. Arc consistency	Define arc consistency. For every value of X_i, there must be a compatible value in X_j satisfying the binary constraint. Explain the AC-3 algorithm or its main steps.	3

5. Example and elimination	Use a suitable CSP example to demonstrate domain reduction and elimination of inconsistent values.	3
6. Efficiency and conclusion	Explain reduced branching, early detection of failure, fewer backtracking steps, and improved search efficiency.	2
Total		15

Consider two variables X and Y, with domains:

$$D(X) = \{1, 2, 3\}, D(Y) = \{2, 3\}$$

Constraint: $X < Y$.

- For $X = 1$, compatible values are $Y = 2$ and $Y = 3$.
- For $X = 2$, the compatible value is $Y = 3$.
- For $X = 3$, no value of Y satisfies $X < Y$; hence remove 3 from $D(X)$.
- After revision, $D(X) = \{1, 2\}$. The domains are consistent with the constraint.

Award marks for a correct example, valid domain revision, and explanation of how propagation reduces search. Equivalent examples are acceptable.

15.

Answer component	**Expected content**	**Marks**
1. Introduction to FOPL	Explain predicates, objects, variables, relations, and the role of quantifiers in representing knowledge.	2
2. Universal quantifier	Explain ∀, meaning “for all.” Give a correct symbolic expression and interpretation.	3
3. Existential quantifier	Explain ∃, meaning “there exists at least one.” Give a correct symbolic expression and interpretation.	3
4. Real-world representations	Translate suitable statements into FOPL using predicates, variables, and appropriate quantifiers.	4

5. Scope and interpretation	Explain the scope of quantifiers, the importance of variable binding, or the distinction between universal and existential statements.	2
6. Conclusion	Explain how quantifiers support formal knowledge representation and logical inference.	1
Total		15

Illustrative examples for valuation
1. Universal quantifier: “All humans are mortal.”

$$\forall x[Human(x) \rightarrow Mortal(x)]$$

This states that every object classified as human is mortal.

2. Existential quantifier: “Some student studies Artificial Intelligence.”

$$\exists x[Student(x) \wedge StudiesAI(x)]$$

This states that at least one student studies Artificial Intelligence.

3. Existential statement with a relationship: “There exists a student who likes mathematics.”

$$\exists x[Student(x) \wedge Likes(x, Mathematics)]$$

Award marks for correct notation, logical connectives, appropriate predicate selection, and accurate interpretation. Equivalent real-world examples are acceptable.

16. Expected answer and mark allocation

Answer component	**Expected content**	**Marks**
1. Introduction to fuzzy logic	Explain the limitations of binary classification and how fuzzy logic represents degrees of membership.	2

2. Linguistic variables	Define a linguistic variable and identify its name, numerical universe, and linguistic terms. Example: performance with Poor, Average, and Excellent.	2
3. Membership functions	Define $\mu_A(x)$, with values from 0 to 1. Explain triangular, trapezoidal, or other suitable membership functions.	3
4. Student-performance example	Illustrate suitable numerical ranges and membership values for Poor, Average, and Excellent. Include a graph, diagram, or numerical example.	3
5. Gradual transitions	Explain overlapping membership functions and partial membership in multiple categories near boundaries.	3
6. Applications and conclusion	Explain how fuzzy representation supports evaluation, decision-making, and handling of imprecise concepts.	2
Total		15

Illustrative example for valuation

Suppose student performance is measured on a scale of 0–100. The following ranges are illustrative; other reasonable membership functions should receive credit.

Example membership functions

Illustrative fuzzy membership functions

Overlapping categories for student performance on a scale from 0 to 100.

